PROMPT ENGINEERING

From AI Basics to Prompt Types, Practical Examples
and Interview Preparation



Clear intent → Better guidance → More useful AI output

1. What Is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computing that builds systems able to perform tasks that normally require human-like intelligence—such as understanding language, recognizing patterns, solving problems, making predictions and generating content.

AI in simple words

AI enables computers to **analyze information and produce useful decisions or outputs** instead of following only fixed instructions.

Generative AI

A type of AI that can **create new content**, such as text, code, images, audio or summaries.

Examples you already know

Application	What AI does
Chatbots	Understand questions and generate natural-language answers.
Recommendation systems	Predict what content or products you may like.
Voice assistants	Recognize speech and respond to requests.
Coding assistants	Generate, explain or improve code.
Image generators	Create images from text instructions.

Interview point

AI is the broader field. An AI model is a system trained to perform particular tasks. Prompting is one way a user communicates with such a model.

2. How Do Humans Communicate With AI?

When we use a generative AI application, we send it an **input**. That input can be a question, instruction, background information, examples, data, or a combination of these.

Step	What happens	Example
1. Human	Decides what they need.	“I need to learn recursion.”
2. Prompt	Writes the request for the AI.	“Explain recursion to a beginner.”
3. AI model	Processes the prompt and predicts a useful response.	Interprets the task and audience.
4. Response	Generates the requested output.	Simple explanation + example.

PROMPT	EXPECTED / SAMPLE OUTPUT
Explain recursion in simple words and give one Python example.	Recursion is a technique where a function calls itself to solve a smaller version of the same problem. **Example:** a countdown function can call itself with 3, then 2, then 1.
Why it works	The prompt specifies both the topic and the level of explanation.

Key idea

The quality of an AI interaction depends partly on **how clearly the user communicates the task**. The prompt acts as the bridge between the user's intention and the model's output.

3. What Is a Prompt?

A **prompt** is the input we give to an AI system to tell it **what we want, what information matters, and sometimes how the answer should look**.

A prompt can contain:

Part	Meaning	Example
Question	Asks for information.	“What is an API?”
Instruction	Tells AI to perform a task.	“Write a Python program.”
Context	Provides useful background.	“I am a beginner.”
Examples	Shows the desired pattern.	“Positive → Good; Negative → Bad.”
Constraints	Sets limits or rules.	“Under 150 words.”
Format	Controls how the answer is presented.	“Use a table.”

What can a prompt ask AI to do?

Explain / Generate

Explain: “Explain REST API to a beginner.”
Generate: “Write Python code for binary search.”

Transform / Analyze

Transform: “Convert this paragraph into a formal email.”
Analyze: “Find the main risks in this plan.”

Simple formula

PROMPT = What we tell the AI to do + information needed to do it well.

4. Prompt, AI Model and Response — Do Not Mix Them Up

Term	Meaning	Example
Prompt	The user's input or instruction.	“Explain Python to a beginner in 5 bullet points.”
AI model	The system that processes the prompt and generates an output.	A language model interprets the request.
Response	The output produced by the model.	Five beginner-friendly bullet points.

Think of it like this:

HUMAN	→	PROMPT	→	AI MODEL	→	RESPONSE
Intent	→	Instructions	→	Processing	→	Generated output

PROMPT	EXPECTED / SAMPLE OUTPUT
Explain Python to a first-year student in 5 bullet points and give 2 real-world examples.	Python is a high-level, readable programming language. • Easy syntax • Large library ecosystem • Used for automation • Used in web development • Popular in data/AI **Examples:** automation scripts; web applications.
Why it works	The model attempts to follow the requested audience, length, topics and examples.

Important

A better prompt can improve relevance and usefulness, but it **does not guarantee factual correctness**. Important information should still be checked.

5. Why Does the Prompt Matter?

A vague prompt forces the AI to guess the audience, depth, scope, examples and format. A specific prompt gives the model fewer important things to guess.

Vague prompt

“Tell me about Python.”

Problem: audience, depth, examples and output format are not specified.

Specific prompt

“Act as a Python teacher. Explain Python to a first-year student in simple English. Cover features, advantages and 3 real-world applications in bullet points.”

Vague	Specific
Tell me about SQL.	Explain SQL JOINs to a beginner with one example for INNER JOIN and LEFT JOIN.
Write code for sorting.	Write Python code for merge sort, show input/output, and give time complexity.
Help with interview.	Act as a junior Python interviewer and ask one DSA question at a time.

Remember

The goal is **not** to make a prompt unnecessarily long. The goal is to include the information that actually matters for the task.

6. What Is Prompt Engineering?

Prompt engineering is the practice of **designing, testing and refining prompts** so an AI model is more likely to produce useful, relevant and task-specific output.

Why do we need it?

Benefit	What it improves
Less ambiguity	Makes the user's goal clearer.
Better control	Can specify audience, tone, length and format.
Consistency	Helps repeated tasks follow the same pattern.
Complex tasks	Breaks requirements into understandable instructions.
Practical use	Useful in coding, education, research, business and interviews.

WRITE	→	TEST	→	EVALUATE	→	REFINE	→	USE
Create		Try		Check output		Improve		Reuse

Interview-ready definition

“Prompt engineering is the process of designing and improving inputs or instructions for AI models to obtain more useful, relevant and consistent outputs.”

7. Anatomy of an Effective Prompt

A practical framework is:

ROLE + TASK + CONTEXT + CONSTRAINTS + OUTPUT FORMAT

You do not always need every part. Use the parts that are relevant to the task.

Part	Question to ask	Example
Role	Who should AI act as?	“Act as a Python teacher.”
Task	What should it do?	“Explain recursion.”
Context	What background matters?	“I am a beginner.”
Constraints	What limits/rules apply?	“Under 200 words.”
Format	How should it look?	“Use bullet points.”
Goal	What should the answer achieve?	“Make it interview-friendly.”

PROMPT	EXPECTED / SAMPLE OUTPUT
Act as a Python teacher. I am a beginner. Explain recursion using a simple analogy and one Python example. Keep it under 200 words and use bullet points.	The model now knows the **role**, **task**, **audience**, **constraints** and **format**.
Why it works	More useful details are supplied without adding irrelevant information.

8. Types of Prompts — First Understand the Map

Prompt types mainly differ in **how much guidance, context, examples, role information or output structure** we give the AI.

Type	Clear definition	Simple example
Zero-Shot	Task is given with no task-specific examples.	“Classify this review as positive or negative.”
One-Shot	One example is shown before the new task.	“Good → Positive. Now classify: Great app.”
Few-Shot	Several examples show the pattern before the new task.	Multiple reviews → labels, then classify a new review.
Role-Based	AI is given a role or perspective to shape its response.	“Act as a technical interviewer.”
Contextual	Relevant background is supplied so AI understands the situation.	“I am a CSE student preparing for placements.”
Instruction-Based	Requirements are stated explicitly.	“Explain JOINs, give examples, stay under 300 words.”
Structured	The required output shape is specified.	“Return a table with Feature, Python, Java.”
Structured Reasoning	The user asks for a clear solution process or checks for a complex task.	“Give approach, complexity, edge cases and code.”
ReAct / Tool Use	AI uses actions/tools, observes results and continues the task.	“Get current weather, then summarize it.”

9. Zero-Shot Prompting

Definition: Zero-shot prompting means asking the AI to perform a task **without giving any task-specific example** inside the prompt. The model relies on the instruction and its learned capabilities.

Easy way to remember

ZERO-SHOT = 0 examples → instruction only.

PROMPT	EXPECTED / SAMPLE OUTPUT
Classify this review as Positive or Negative: "The app is very easy to use."	**Positive**
Why it works	The task is clear enough that an example is not necessary.

When should you use it?

• When the task is simple and the desired behavior is obvious.

• When you want a short prompt with minimal setup.

• For common tasks such as summarization, translation or basic classification.

Another example

Prompt: "Translate this sentence into Hindi: 'I have an interview tomorrow.'"
Output: "■■■■ ■■ ■■■■■■■■ ■■■"

Interview phrase

"Zero-shot prompting performs a task without providing task-specific examples in the prompt."

10. One-Shot Prompting

Definition: One-shot prompting gives the AI **exactly one example** showing how an input should be handled. The model then applies the demonstrated pattern to a new input.

Easy way to remember

ONE-SHOT = 1 example → example + new task.

PROMPT	EXPECTED / SAMPLE OUTPUT
Example: "I love this phone." → Positive **New: "The battery life is terrible." → ?**	**Negative**
Why it works	The first example teaches the expected classification pattern.

Why is the example useful?

• It can clarify an output label or format.

• It reduces uncertainty when the expected pattern is not obvious.

• It is useful when one demonstration is enough to show the rule.

Real-world example

Example: "Meeting moved to 4 PM." → Formal: "The meeting has been rescheduled to 4 PM."
New: "Send the report by Friday." → The model can produce a similar formal version.

Interview phrase

"One-shot prompting uses a single demonstration inside the prompt to guide the model."

11. Few-Shot Prompting

Definition: Few-shot prompting provides **multiple examples** so the AI can infer a pattern, style, classification rule or output format before handling a new input.

Easy way to remember

FEW-SHOT = multiple examples → examples + new input.

PROMPT	EXPECTED / SAMPLE OUTPUT
“Great service!” → Positive **“Worst experience.” → Negative** **“Very helpful staff.” → Positive** **New: “The staff were rude.” → ?**	**Negative**
Why it works	Several examples make the positive/negative pattern clear.

Where it helps

Specialized classification

Useful when the task uses labels or rules that are specific to your project.

Consistent style / format

Useful when you want outputs to follow the same structure as your examples.

Important

Few-shot prompting is **not retraining the model**. The examples are provided only as part of the current prompt.

12. Role-Based / Persona Prompting

Definition: Role-based prompting tells the AI to respond from a particular **profession, role or perspective**. This can influence vocabulary, tone, priorities and the type of guidance it gives.

Easy way to remember

ROLE = Who should the AI act like?

PROMPT	EXPECTED / SAMPLE OUTPUT
Act as a technical interviewer for a Python developer role. Ask me one DSA question at a time.	The AI behaves like an interviewer: it asks questions, waits for the candidate's answer and can evaluate the response.
Why it works	The role provides a behavioral frame for the conversation.

Other examples

Role	Example prompt
Teacher	“Act as a Python teacher and explain lists to a beginner.”
Recruiter	“Act as a technical recruiter and review my resume for a Django role.”
Career coach	“Act as a career coach and help me improve my interview answers.”
Customer-support agent	“Act as a support agent and explain the troubleshooting steps politely.”

Interview phrase

“Role prompting gives the model a perspective or behavioral frame for its response.”

13. Contextual Prompting

Definition: Contextual prompting supplies **relevant background information** so the AI understands the user's situation, audience, purpose or constraints.

Easy way to remember

CONTEXT = What does the AI need to know about the situation?

PROMPT	EXPECTED / SAMPLE OUTPUT
I am a CSE student preparing for placements. Explain REST API in simple English and give 3 interview questions.	The answer can be beginner-friendly, placement-oriented and include interview-focused questions.
Why it works	Without the context, the model may give a more general or advanced explanation.

What can context include?

• Your experience level: beginner, intermediate or advanced.

• Your goal: exam preparation, interview, project work, presentation, etc.

• Your audience: students, manager, customer, technical team.

• Relevant background data: project details, constraints or previous decisions.

Interview phrase

“Context reduces missing information about the user's situation and goal.”

14. Instruction-Based Prompting

Definition: Instruction-based prompting states **exactly what the AI should do** and clearly lists important requirements, rules or constraints.

Easy way to remember

INSTRUCTION = What exactly should the AI do?

PROMPT	EXPECTED / SAMPLE OUTPUT
Explain SQL JOINs to a beginner. Cover INNER, LEFT and RIGHT JOINs, give one example for each, and keep it under 300 words.	The response should cover the three JOIN types, include one example for each and stay within the requested length.
Why it works	The task and requirements are explicit, so the AI has less to guess.

Weak vs strong instruction

Weak	Strong
“Explain SQL.”	“Explain SQL JOINs to a beginner, compare INNER and LEFT JOINs, and give one table example.”
“Write code.”	“Write Python code for binary search, include input/output and time complexity.”

Interview phrase

“Instruction prompting focuses on clearly specifying what the model should do.”

15. Structured Prompting

Definition: Structured prompting tells the AI **how the final answer should be organized**. It controls the shape or format of the output, not just the topic.

Easy way to remember

STRUCTURED = What should the output look like?

PROMPT	EXPECTED / SAMPLE OUTPUT
Compare Python and Java using a table with columns: Feature, Python, Java and Best Use.	**Feature | Python | Java | Best Use** Syntax | Simple | Verbose | Python for quick development; Java for large enterprise systems.
Why it works	The requested columns make the output easier to read, compare and reuse.

Common output formats

Format	Useful for
Bullet points	Quick notes, revision and summaries.
Table	Comparisons and structured data.
Numbered steps	Tutorials and procedures.
JSON / key-value	Machine-readable or repeatable workflows.
Template	Emails, reports, resumes and documentation.

Interview phrase

“Structured prompts control the shape of the output, not only its content.”

16. Structured Reasoning

Definition: For complex tasks, structured reasoning prompting asks the AI to provide an **understandable solution process, checks or intermediate results** rather than only the final answer.

Important distinction

Ask for useful, user-visible steps such as **approach** → **checks** → **complexity** → **final answer**. You do not need to request hidden/private chain-of-thought.

PROMPT	EXPECTED / SAMPLE OUTPUT
Solve this DSA problem. Give the approach, explain the key steps, mention edge cases, give time/space complexity and then provide the final code.	**Approach:** explain the algorithm. **Checks:** discuss edge cases. **Complexity:** give time and space. **Final:** provide code.
Why it works	The requested structure makes a complex solution easier to study and verify.

Useful for

DSA / Mathematics

Break a problem into a clear method, calculations and checks.

Debugging / Planning

List likely causes, diagnostic steps, decisions and the final recommendation.

Interview phrase

“Structured reasoning asks for an understandable solution process and checks for multi-step tasks.”

17. ReAct / Tool-Using Workflow

Definition: In a tool-using workflow, an AI system can decide what action is needed, **use a tool**, observe the result and continue until it can answer the task.

Stage	What happens	Example
1. Need	Identify information or action required.	User asks for current weather.
2. Plan	Determine which tool/action can help.	Use a weather tool.
3. Action	Call the tool.	Fetch current weather data.
4. Observe	Read the returned result.	Temperature, rain chance, wind.
5. Respond	Use the result to answer.	Give a concise weather summary.

Where it helps

Current web information, calculations, database queries, code execution, file analysis, APIs and other external actions.

PROMPT	EXPECTED / SAMPLE OUTPUT
Find the current weather for Indore, then tell me whether I should carry an umbrella.	The system needs current weather data first; after checking the result, it can give a recommendation based on rain conditions.
Why it works	The key idea is **reason/plan** → **act/use tool** → **observe** → **continue**.

Interview phrase

“ReAct is commonly described as combining reasoning with actions in a tool-using workflow.”

18. Zero-Shot vs One-Shot vs Few-Shot

These three are easiest to remember by **counting the number of examples** supplied in the prompt.

Type	Examples given	What AI receives	Best use
Zero-Shot	0	Instruction only	Direct / common tasks
One-Shot	1	Instruction + one demonstration	Clarify one pattern or format
Few-Shot	Multiple	Instruction + several demonstrations	Specialized patterns, style or classification

ZERO-SHOT

0 examples

“Classify this review.”

ONE-SHOT

1 example

“Good → Positive. Now classify this.”

FEW-SHOT

Multiple examples

“Great → Positive; Bad → Negative; Helpful → Positive. Now classify this new review.”

Memory trick

0 → Zero-shot 1 → One-shot Many → Few-shot

19. One Prompt Can Combine Multiple Techniques

Real-world prompts often combine several techniques. The types are **not mutually exclusive**; combining them can make a prompt more useful.

PROMPT	EXPECTED / SAMPLE OUTPUT
Act as a senior technical interviewer. I am a CSE student preparing for a Python/Django placement interview. Ask one question at a time. Start with medium-level DSA, then move to Python and Django. After each answer, evaluate my approach and give a short improvement tip. Do not reveal the answer before I attempt it.	**Role-Based:** senior interviewer **Contextual:** CSE student + placement goal **Instruction-Based:** ask one question at a time **Constraints:** do not reveal answer early **Structured interaction:** question → answer → feedback → next
Why it works	A strong real-world prompt often combines multiple useful elements.

Technique	What it adds
Role-Based	A perspective or professional behavior.
Contextual	Background about the user and situation.
Instruction-Based	Specific actions and requirements.
Constraints	Rules such as length, order or what not to do.
Structured output	A predictable response format or interaction flow.

Key takeaway

Good prompt engineering means choosing the **right combination of guidance** for the task—not using every technique every time.

20. Common Interview Questions — With Clear Answers

Question	Short, interview-ready answer
What is a prompt?	A prompt is the input or instruction given to an AI model to guide the output.
What is prompt engineering?	It is the process of designing, testing and refining prompts to obtain more useful, relevant and consistent outputs.
What is zero-shot prompting?	Performing a task without providing task-specific examples in the prompt.
One-shot vs few-shot?	One-shot gives one example; few-shot gives multiple examples.
Why is context important?	It gives relevant background such as audience, situation, goal or constraints.
Does a longer prompt always work better?	No. Relevance and clarity matter more than length. Unnecessary details can make the request harder to follow.
Can prompt engineering guarantee correct answers?	No. AI can still make mistakes or unsupported claims, so important output should be verified.
What is an AI hallucination?	A hallucination is when an AI produces information that sounds plausible but is false, unsupported or incorrectly invented.

Good interview line for hallucinations

“Prompt engineering can improve clarity and reduce some errors, but it cannot guarantee truth. Important facts should be verified against reliable sources.”

21. Final Revision — Remember This Flow

Concept	Remember it as
AI	Broader field of building intelligent systems.
Prompt	Input / instruction given to an AI system.
Prompt engineering	Design → test → evaluate → refine prompts.
Zero-shot	0 examples.
One-shot	1 example.
Few-shot	Multiple examples.
Role-Based	Adds a role or perspective.
Contextual	Adds relevant background.
Instruction-Based	States exactly what to do.
Structured	Controls output format.
Structured reasoning	Requests useful solution steps/checks.
ReAct / Tool Use	Plan → action/tool → observe → respond.

Strong prompt checklist

Before sending a prompt, ask: **Is my goal clear? Is the audience clear? Did I provide relevant context? Did I state important constraints? Did I specify the desired format?**

Interview one-liner

“Prompt engineering is the practice of communicating our intent to an AI model in a clear, structured and task-specific way so that the generated output better matches our goal.”

22. Conclusion

Prompt engineering is a practical skill for working effectively with generative AI. The aim is not to memorize dozens of prompt templates. The real skill is understanding **what information the AI needs to perform the task well**.

1	CLEAR INTENT
2	BETTER PROMPT
3	BETTER GUIDANCE
4	MORE USEFUL OUTPUT

The 6 things to remember

• **Role** — who should the AI act as?

• **Task** — what exactly should it do?

• **Context** — what background is relevant?

• **Constraints** — what rules, limits or conditions apply?

• **Examples** — is a demonstration needed?

• **Format** — how should the final answer look?

Final classroom line

“AI gives us the capability; prompt engineering helps us communicate our intent clearly.”

Thank you — Questions?